

Pilot Validation Report

Synthetic SSR Pipeline vs. Real Human Responses

Generated: April 30, 2026

Sample

Human respondents: **33**

Age: 22-26: 32, 26-30: 1

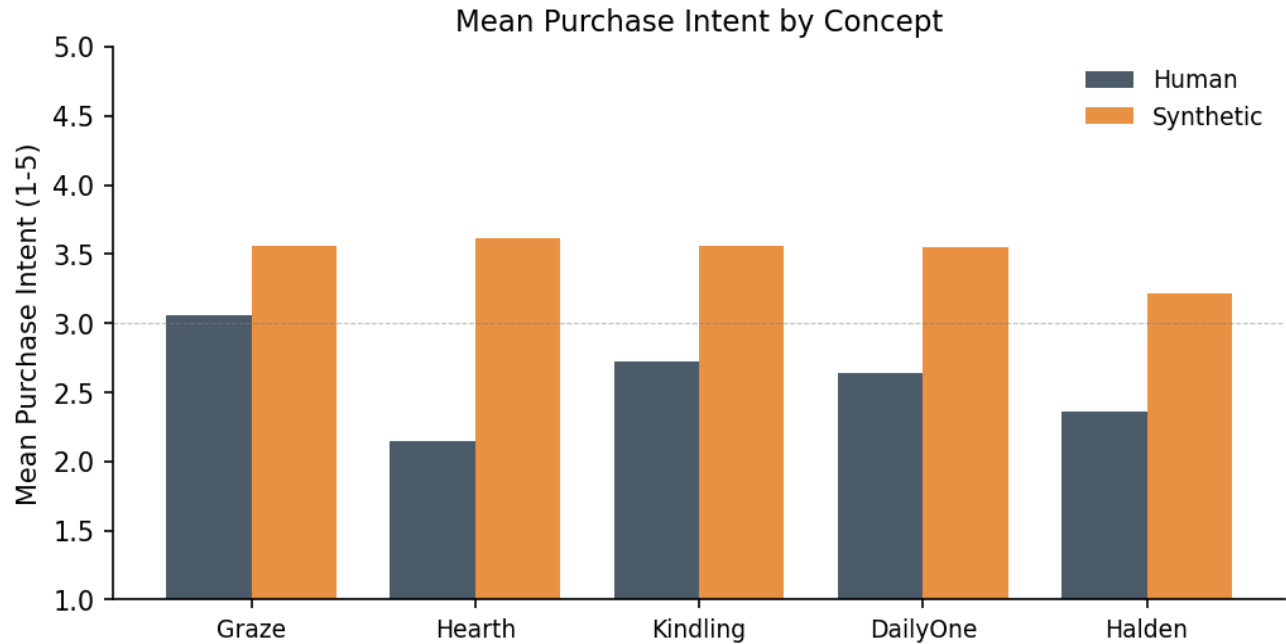
Status: Between things: 1, College student: 2, Grad student: 5, Working full time: 25

Synthetic personas per concept: **100**

What this report can and cannot tell you

This is a **pilot validation**. With 5 concepts, statistical correlations have wide confidence intervals and should not be reported as precise figures. Read the ranking agreement and mean-tracking comparisons as directional evidence — they show whether the pipeline is broadly tracking human responses, not whether it has a specific accuracy %. Distributional and ranking concordance on this small N is a sanity check, not a benchmark.

Top-line summary



Concept	Human Mean	Synth Mean	Δ	Human Top-2-Box	Synth Top-2-Box
Graze	3.06	3.56	+0.50	39%	54%
Hearth	2.15	3.62	+1.46	18%	55%
Kindling	2.73	3.56	+0.83	30%	56%
DailyOne	2.64	3.55	+0.91	21%	58%
Halden	2.36	3.22	+0.85	18%	43%

Ranking agreement

Human ranking (best $\rightarrow$ worst): **Graze** > **Kindling** > **DailyOne** > **Halden** > **Hearth**

Synthetic ranking (best $\rightarrow$ worst): **Hearth** > **Kindling** > **Graze** > **DailyOne** > **Halden**

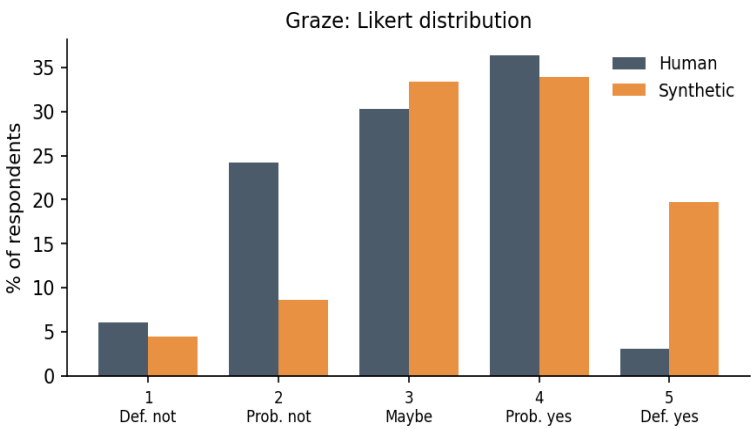
Spearman rank correlation: $\rho = -0.10$

Best-concept agreement: human picked **Graze**, synthetic picked **Hearth** ✗

Worst-concept agreement: human picked **Hearth**, synthetic picked **Halden** ✗

Spearman $\rho = +1.0$ means perfect ranking agreement; 0 means random; -1.0 means inverted. With $n=5$ concepts, $\rho \geq 0.80$ is a strong directional signal but the confidence interval is wide.

Graze



Metric	Human	Synthetic
Mean PI	3.06	3.5
Std	1.00	0.2
Top-2-Box %	39%	54%
Bottom-2-Box %	30%	13%
KS distance	—	0.1
n	33	10

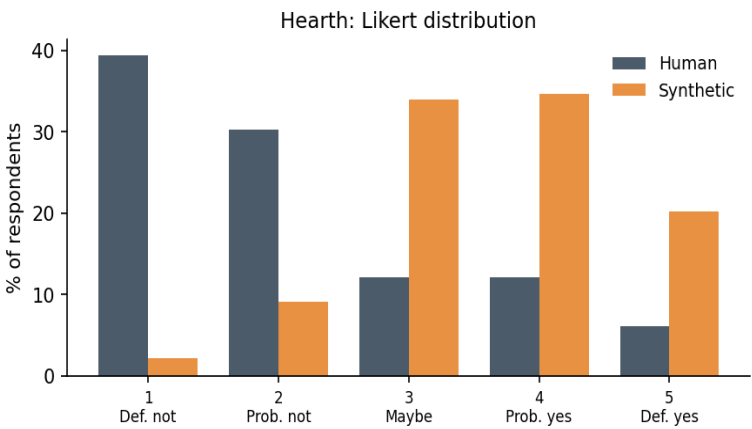
Sample human responses

“I’m interested in incorporating more protein into my diet and I also really enjoy ice cream. I think if it came in a different flavor or was slightly cheaper I may be more inclined to purchase it.”

“The savory thing is throwing me and I don’t rly eat ice cream a lot”

“protein aspect, I love miso (interesting flavor) and its clean branding”

Hearth

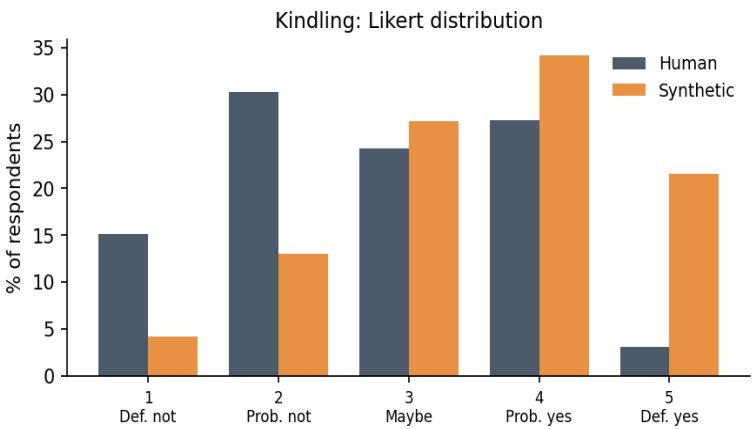


Metric	Human	Synthetic
Mean PI	2.15	3.6
Std	1.25	0.1
Top-2-Box %	18%	55%
Bottom-2-Box %	70%	11%
KS distance	—	0.5
n	33	10

Sample human responses

- “Ashwaganda drink i don’t know enough about ashwaganda. Nor do I want a calming drink”
- “Seems holistic / fake”
- “not a fan of carbonation”

Kindling

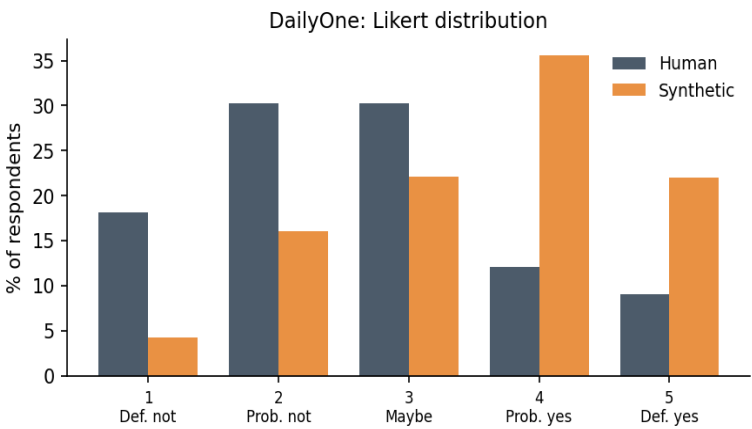


Metric	Human	Synthetic
Mean PI	2.73	3.5
Std	1.13	0.3
Top-2-Box %	30%	56%
Bottom-2-Box %	45%	17%
KS distance	—	0.2
n	33	10

Sample human responses

- “I don’t drink coffee. Otherwise it seems good, a little fancy for my tastes.”
- “Not big fan of coffee in the first place”
- “Had a bad experience at the Addis Ababa airport. I have free coffee at work”

DailyOne

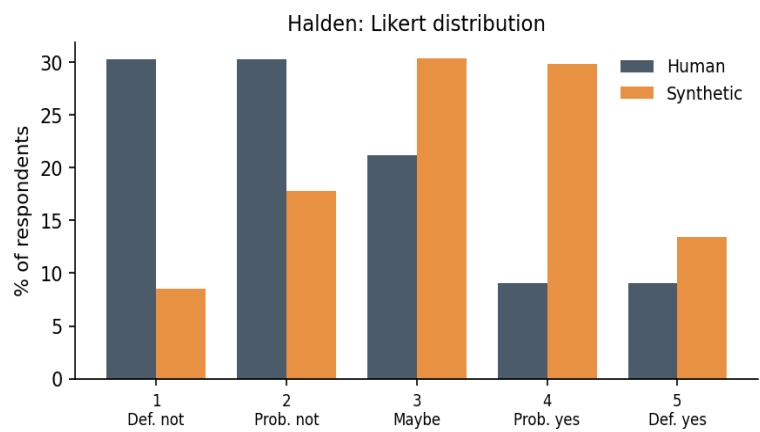


Metric	Human	Synthetic
Mean PI	2.64	3.5
Std	1.19	0.1
Top-2-Box %	21%	58%
Bottom-2-Box %	48%	20%
KS distance	—	0.3
n	33	10

Sample human responses

- “Don’t use multi vitamins”
- “Neutral”
- “How is this different than other multivitamins”

Halden



Metric	Human	Synthetic
Mean PI	2.36	3.2
Std	1.27	0.4
Top-2-Box %	18%	43%
Bottom-2-Box %	61%	26%
KS distance	—	0.3
n	33	9

Sample human responses

- “It’s too expensive and not really my style. Too stylish.”
- “Not my style”
- “not my style and it’s expensive”

Methodology notes

Likert ratings collected via Google Forms (33 respondents). Concept order in the form: Graze, Hearth, Kindling, DailyOne, Halden. Likert label mapping: 'Definitely would not buy' = 1 through 'Definitely would buy' = 5.

Synthetic results from the SSR pipeline. Per-respondent expected ratings ($E[r]$) computed by averaging PMFs across 6 reference sets. Per-concept aggregate PMFs computed by averaging individual PMFs. KS distance is the max absolute difference between cumulative distributions (0 = identical, 1 = no overlap).

Limitations: small concept sample ($n=5$) makes precise correlation statistics unreliable. Ranking agreement and directional mean tracking are the primary signals. Human respondents were a friend/follower convenience sample — not a representative consumer panel.